

Hariprakash Karthikeyan

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SUMMARY

Data Scientist who builds production-oriented data and ML systems for messy, high-stakes problems. Owned pipeline design, feature engineering, and modeling across healthcare, marketing science, and renewable energy, prioritizing correctness, interpretability, and system reliability over vanity metrics.

EDUCATION

University of Rochester

MS, Data Science

Rochester, New York

Aug 2024 – Dec 2025

• **Relevant Coursework:** Data Science at Scale, Statistical Machine Learning, Network Science Analytics, Database Systems

SRM Institute of Science and Technology

Bachelor of Technology, Electronics and Communication Engineering

Chennai, India

Aug 2020 – May 2024

SKILLS

Programming & Querying: Python, SQL, R

Data Analysis & ML: Pandas, NumPy, Scikit-learn, Feature Engineering, Time Series Analysis

Deep Learning: TensorFlow, Keras, PyTorch, CNNs, Autoencoders, VAEs

Data Systems: SQL Databases, MongoDB, Spark, Databricks

Visualization & BI: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Cloud & Tools: AWS, Azure, Git, Docker, Streamlit, Linux

EXPERIENCE

University of Rochester Medical Center

Data Scientist / Research Assistant

Rochester, New York

Aug 2025 – Dec 2025

- Engineered a patient-level, time-aligned EHR dataset integrating labs, vitals, and echocardiographic **data across 2,000+ patients**, with focused analysis on a curated cohort of 845 patients spanning multiple hospitalizations
- Built a rule-based NLP pipeline to extract clinically meaningful cardiac phenotypes (LVEF, wall motion, valvular, systolic and diastolic dysfunction) **from 2–3+ echo reports per patient**, enabling structured analysis of narrative imaging data
- Designed a reusable, **time-series-ready data foundation** preserving original measurement timestamps and hospitalization context; outputs were reviewed and validated by PIs and clinicians for downstream modeling and longitudinal analysis

Butler/Till

Data Scientist

Rochester, New York

Aug 2025 – Dec 2025

- Led feature engineering and hierarchical Bayesian modeling** for a geo-level Media Mix Model, extending prior HBM/MMM research to capture carryover (adstock), saturation (Hill functions), and geographic heterogeneity across ~170 DMAs
- Built a scalable Bayesian MMM pipeline on a weekly **DMA panel of ~3,000–4,000 observations**, integrating demographic context, nonlinear media transformations, and partial pooling to stabilize estimates in sparse markets
- Delivered model outputs and insights to Butler/Till sponsors, demonstrating **improved interpretability and reliability over national-level models**, and supporting region-specific ROI estimation and scenario analysis

Buffalo Solar

Data Analyst Intern

Buffalo, New York

June 2025 – July 2025

- Designed and deployed an end-to-end analytics system integrating SolarEdge APIs and CPS Excel data to **monitor 468 solar sites**, replacing fragmented, manual data checks with a single source of operational truth
- Built a **production Streamlit + Plotly dashboard** demoed weekly across the company, enabling real-time outage detection, multi-site comparisons, and predicted vs. actual energy analysis that surfaced previously unseen energy loss
- Owned the full data lifecycle** from ingestion and defensive parsing to visualization and export, introducing configurable thresholds and scalable site loading that eliminated manual workflows and drastically reduced analysis effort across teams

PROJECTS

End-to-End Distributed Sentiment Analytics (Databricks, Spark)

- Built a production-style sentiment analytics pipeline on Databricks using **Apache Spark and Delta Lake**, transforming raw model outputs into analytics-ready Gold tables across **100k+ text records**
- Evaluated pipeline performance using Spark UI** (spill, skew, shuffle, storage, serialization) and logged model metrics and artifacts with MLflow, demonstrating efficient execution and reproducible analysis

NYC Taxi & Ride-Hailing Network Analysis (Large-Scale Graph Modeling)

- Constructed spatiotemporal directed mobility networks from **~60GB of NYC TLC taxi and FHV trip data**, applying network centrality measures and panel regressions to analyze congestion, fare dynamics, and structural bottlenecks